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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001
B. Sc DEGREE EXAMINATION APRIL 2021
BC304S – ANALYTICAL BIOCHEMISTRY

Time: 3 hrs **Max Marks: 75**

SECTION – A (25X1=25)
Answer ALL the questions

Choose the correct answer:

1. The principal cation in extracellular fluid is ----------
   a) Sodium b) Potassium c) Calcium d) Magnesium
2. The normal pH of blood is ------
   a) 7.0 b) 7.1 c) 7.2 d) 7.4
3. The internal resistance offered by a liquid or a gas to flow is called --------
   a) Surface tension b) osmosis c) viscosity d) adsorption
4. The instrument used for measurement of coloured substances is ----------
   a) calorimeter b) colorimeter c)coulorimeter d) sonicator
5. ---------- centrifugation depends only upon the buoyant density of the particle.
   a)Rate zonal b) Isopycnic c) Differential d) None
6. The basic unit of sedimentation coefficient is ---------
   a)5 x 10^{-13} sec b) 3 x 10^{-13} sec c) 1 x 10^{-14} sec d) 1 x 10^{-13} sec
7. The phenomenon in which compounds exhibit fluorescence naturally is called ------- fluorescence.
   a)intrinsic b) extrinsic c)quenching d)adsorption

8. ---------- imparts considerable buffering power to haemoglobin near pH 7.

   a)Histidine b)Proline c)Leucine d)Lysine

9. ---------- disperses the radiation into its component wavelengths.

   a) Entrance slit b) Collimating lens c) Grating d) Exit slit

10. The radiation source in atomic absorption spectroscopy is ------ lamp.

    a) Hydrogen b) Deuterium c) Hollow cathode d) Fluorescent

Say true or false:

11. Erythrocytes are isotonic with plasma fluid.
12. The most commonly used pH dependent unit is the glass electrode.
13. Io/I is known as transmittance.
14. Only one monochromator is used in spectrofluorimetry.
15. Analytical ultracentrifugation detects conformational changes in DNA and protein.

Answer in one or two sentences:

16. Define sedimentation rate
17. What is osmosis?
18. Differentiate intrinsic and extrinsic fluorescence
19. What is buffer capacity?
20. State Beer-Lambert's law.
21. Write a note on viscosity.
22. Differentiate wavelength and wave number.
23. What is a reference electrode? Give an example.

24. Define fluors.
25. What are the characteristics of a good gradient material?

SECTION – B (4X5=20)

Answer any FOUR questions:

26. Give a brief account on biologically important buffers.
27. Describe spectrofluorimetry briefly
28. Explain the construction and application of a glass electrode.
29. Illustrate the different types of rotors.
30. Explain the principle and applications of IR spectrophotometry.
31. Write a note on analytical ultracentrifugation.

SECTION – C (3X10=30)

Answer any THREE questions

32. Explain density gradient centrifugation in detail.
33. Describe the principle, construction and applications of UV-Visible spectrophotometry.
34. Write a note on Clarke's oxygen electrode.
35. Explain the principle, technique and applications of atomic absorption spectroscopy.
36. What is surface tension? Explain the factors affecting surface tension.

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